

EXP - 18

Code:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
int main()
{
int n,i,j;
char a[50][50];
printf("enter the no: intermediate code:");
scanf("%d",&n);
for(i=0;i<n;i++)
{
printf("enter the 3 address code:%d:",i+1);
for(j=0;j<6;j++)
{
scanf("%c",&a[i][j]);
}
}
printf("the generated code is:");
for(i=0;i<n;i++)
{
printf("\n mov %c,R%d",a[i][3],i);
if(a[i][4]=='-')
{
printf("\n sub %c,R%d",a[i][5],i);
}
}
```

```

if(a[i][4]=='+')
{
printf("\n add %c,R%d",a[i][5],i);
}
if(a[i][4]=='*')
{
printf("\n mul %c,R%d",a[i][5],i);
}
if(a[i][4]=='/')
{
printf("\n div %c,R%d",a[i][5],i);

}
printf("\n mov R%d,%c",i,a[i][1]);
printf("\n");
}
return 0;
}

```

Output:

```

enter the no: intermediate code:2
enter the 3 address code:1:a=b+c
enter the 3 address code:2:d=n*d
the generated code is:
    mov b,R0
    add c,R0
    mov R0,a

    mov n,R1
    mul d,R1
    mov R1,d

```